

Database Systems Lab

Lab Task 2

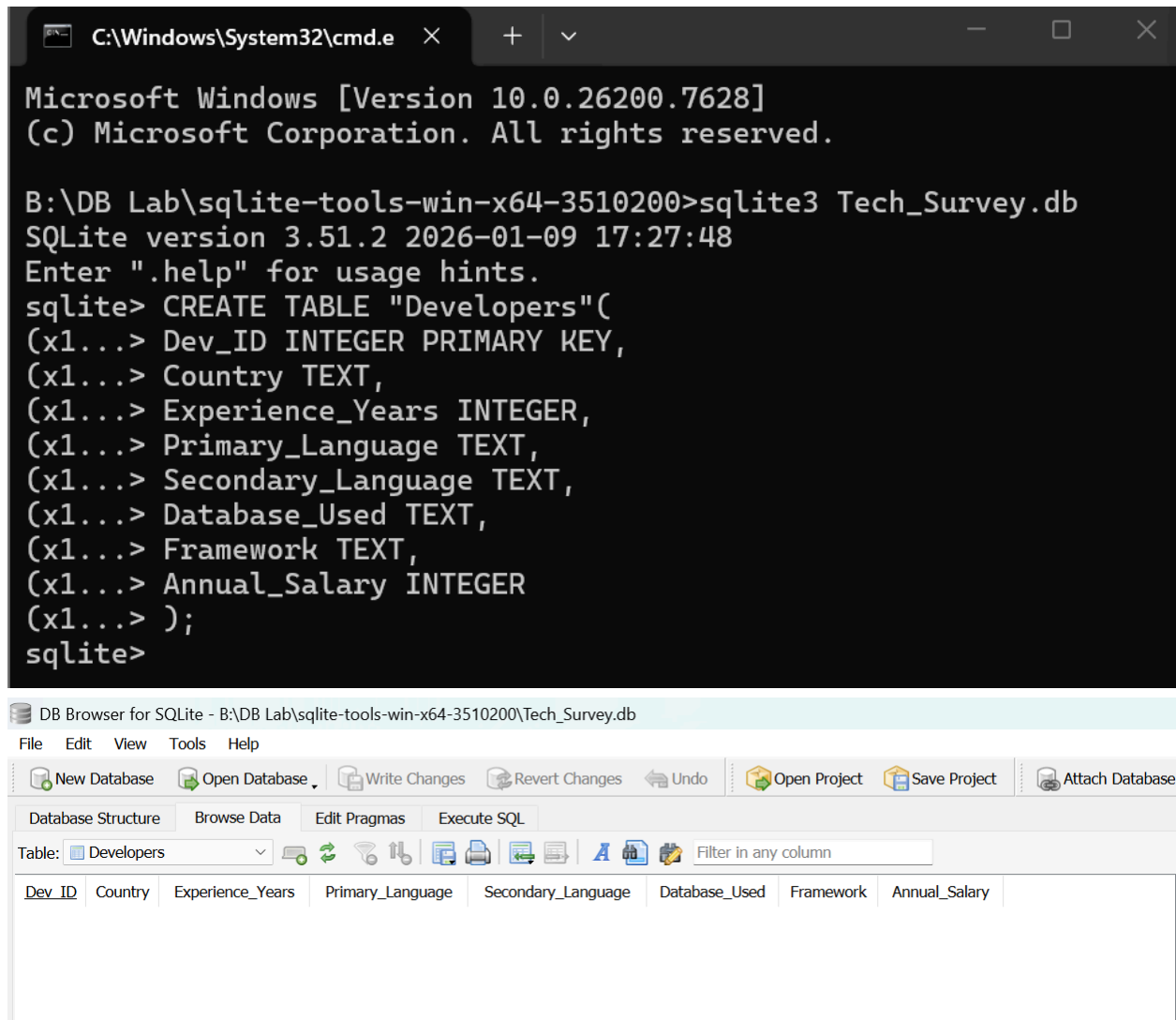


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24P-0554

Section 4C

Operation 0



The image shows a Windows command prompt window and a DB Browser for SQLite application. The command prompt window is titled "C:\Windows\System32\cmd.e" and displays the following text:

```
Microsoft Windows [Version 10.0.26200.7628]
(c) Microsoft Corporation. All rights reserved.

B:\DB Lab\sqlite-tools-win-x64-3510200>sqlite3 Tech_Survey.db
SQLite version 3.51.2 2026-01-09 17:27:48
Enter ".help" for usage hints.
sqlite> CREATE TABLE "Developers"(
(x1...> Dev_ID INTEGER PRIMARY KEY,
(x1...> Country TEXT,
(x1...> Experience_Years INTEGER,
(x1...> Primary_Language TEXT,
(x1...> Secondary_Language TEXT,
(x1...> Database_Used TEXT,
(x1...> Framework TEXT,
(x1...> Annual_Salary INTEGER
(x1...> );
sqlite>
```

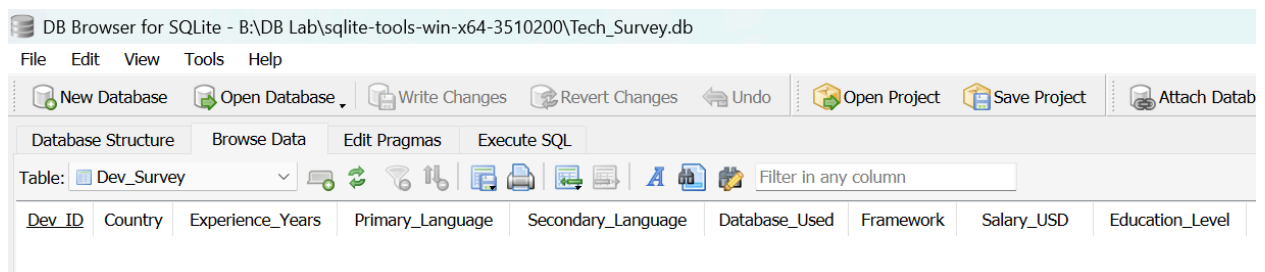
The DB Browser for SQLite application is titled "DB Browser for SQLite - B:\DB Lab\sqlite-tools-win-x64-3510200\Tech_Survey.db". It has a menu bar (File, Edit, View, Tools, Help) and a toolbar with buttons for New Database, Open Database, Write Changes, Revert Changes, Undo, Open Project, Save Project, and Attach Database. The interface shows the "Database Structure" tab selected, with a table named "Developers" selected. The table structure is displayed as follows:

Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Annual_Salary
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Operation 1

ALTER TABLE "Dev_Survey"
ADD COLUMN "Education_Level" TEXT

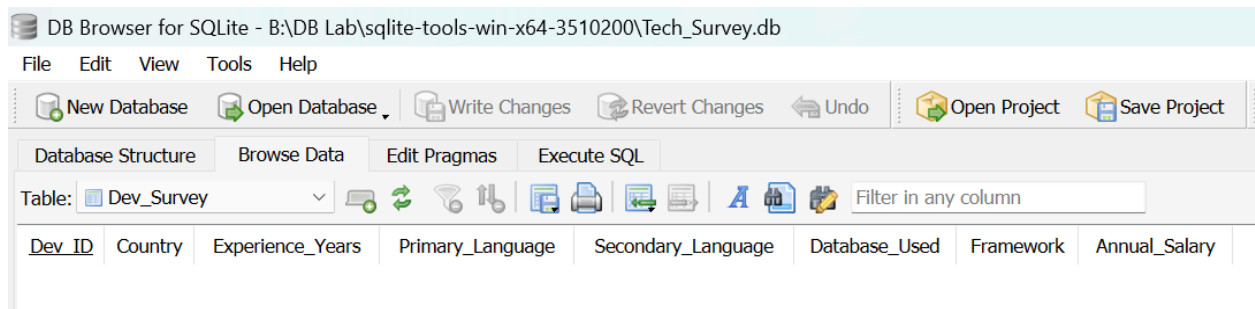
```
sqlite> ALTER TABLE "Dev_Survey"  
...> ADD COLUMN "Education_Level" TEXT;  
sqlite>
```



Operation 2

ALTER TABLE "Developers"
RENAME TO "Dev_Survey"

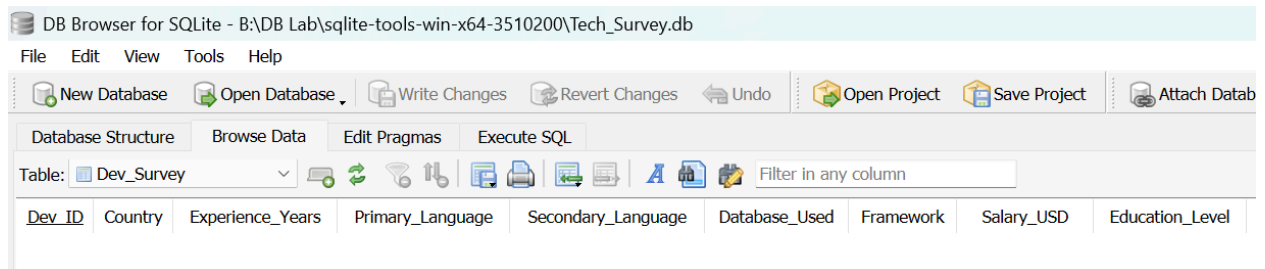
```
(x1...> Annual_Salary INTEGER  
(x1...> );  
sqlite> ALTER TABLE "Developers"  
...> RENAME TO "Dev_Survey";  
sqlite>
```



Operation 3

ALTER TABLE "Dev_Survey"
RENAME "Annual_Salary" TO "Salary_USD"

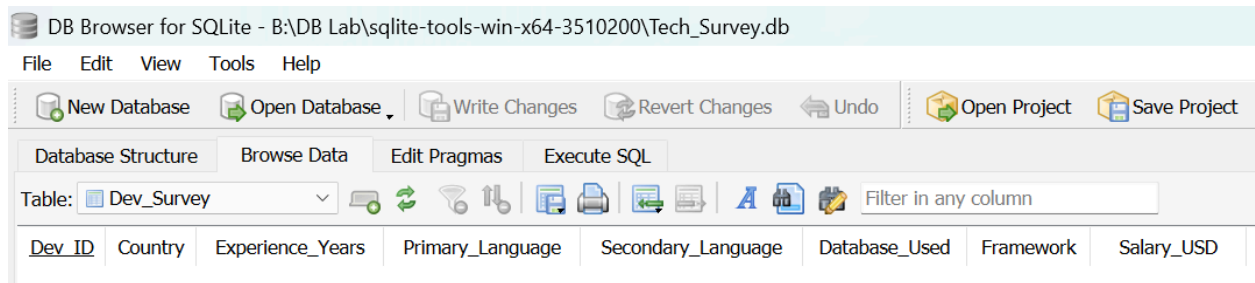
```
Parse error: no such table: Developers  
sqlite> ALTER TABLE "Dev_Survey"  
...> RENAME "Annual_Salary" TO "Salary_USD";  
sqlite>
```



Operation 4

ALTER TABLE "Dev_Survey"
DROP COLUMN "Education_Level"

```
sqlite> ALTER TABLE "Dev_Survey"  
...> DROP COLUMN "Education_Level";  
sqlite>
```



Operation 5

INSERT INTO "Dev_Survey"
VALUES(01,'Pakistan',10,'Urdu','English','My
Sql','DB',50000);

```
sqlite> INSERT INTO "Dev_Survey"  
...> VALUES(01, 'Pakistan', 10, 'Urdu', 'English', 'MySQL', 'DB', 50000);  
sqlite>
```

Database Structure Browse Data Edit Pragmas Execute SQL							
Table: Dev_Survey							
Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1 Paki...	10	Urdu	English	MySQL	DB	50000

Operation 6

INSERT INTO 'Dev_Survey'
("Dev_ID",Experience_Years,"Framework",Sa
lary_USD)
VALUES(02,15,'MySqlLite',60000);

```
sqlite> INSERT INTO "Dev_Survey" ("Dev_ID",Experience_Years,"Framework",Salary_USD)
...> VALUES(02,15,'MySqlLite',60000);
sqlite>
```

Database Structure Browse Data Edit Pragma's Execute SQL								
Table: Dev_Survey								
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Paki...	10	Urdu	English	MySQL	DB	50000
2	2	NULL	15	NULL	NULL	NULL	MySqlLi...	60000

Operation 7

INSERT INTO "Dev_Survey"
VALUES

(03,'USA',12,'English','Spanish',MySq','CN',7
0000)

(04,'Pakistan',3,'C','C++',Gemini','RSD',8000
0)

(05,'UAE',7,'C_Sharp','Rust',Grok','PUT',900
00)

```
sqlite> INSERT INTO "Dev_Survey"  
...> VALUES  
...> (03,'USA',12,'English','Spanish','MySQ','CN',70000),  
...> (04,'Pakistan',3,'C','C++','Gemini','RSD',80000),  
...> (05,'UAE',7,'C_Sharp','Rust','Grok','PUT',90000);  
sqlite>
```

Table: Dev_Survey								
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Paki...	10	Urdu	English	MySQL	DB	50000
2	2	NULL	15	NULL	NULL	NULL	MySQLLi...	60000
3	3	USA	12	English	Spanish	MySQL	CN	70000
4	4	Paki...	3	C	C++	Gemini	RSD	80000
5	5	UAE	7	C_Sharp	Rust	Grok	PUT	90000

Operation 8

UPDATE "Dev_Survey"
SET Salary_USD = Salary_USD +
(Salary_USD * 0.01)
WHERE Country = 'Pakistan';

```
sqlite> UPDATE "Dev_Survey"  
...> SET Salary_USD = Salary_USD + (Salary_USD * 0.10)  
...> WHERE Country = 'Pakistan';  
sqlite>
```

Database Structure Browse Data Edit Pragmas Execute SQL								
Table: Dev_Survey Filter in any column								
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Paki...	10	Urdu	English	MySQL	DB	55000
2	2	NULL	15	NULL	NULL	NULL	MySQLLi...	60000
3	3	USA	12	English	Spanish	MySQL	CN	70000
4	4	Paki...	3	C	C++	Gemini	RSD	88000
5	5	UAE	7	C_Sharp	Rust	Grok	PUT	90000

Operation 9

UPDATE "Dev_Survey"
SET Framework = NULL
WHERE Database_Used IS NULL;

```
sqlite> UPDATE "Dev_Survey"  
...> SET Framework = NULL  
...> WHERE Database_Used IS NULL;  
sqlite>
```

Database Structure Browse Data Edit Pragma's Execute SQL								
Table: Dev_Survey								
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Paki...	10	Urdu	English	MySQL	DB	55000
2	2	NULL	15	NULL	NULL	NULL	NULL	60000
3	3	USA	12	English	Spanish	MySQL	CN	70000
4	4	Paki...	3	C	C++	Gemini	RSD	88000
5	5	UAE	7	C_Sharp	Rust	Grok	PUT	90000

Operation 10

UPDATE "Dev_Survey"
SET "Primary_Language" = 'Python'
WHERE Dev_ID = 5;

```
sqlite> UPDATE "Dev_Survey"  
...> SET "Primary_Language" = 'Python'  
...> WHERE Dev_ID = 5;  
sqlite>
```

Database Structure Browse Data Edit Pragmas Execute SQL								
Table: Dev_Survey Filter in any column								
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Paki...	10	Urdu	English	MySQL	DB	55000
2	2	NULL	15	NULL	NULL	NULL	NULL	60000
3	3	USA	12	English	Spanish	MySQL	CN	70000
4	4	Paki...	3	C	C++	Gemini	RSD	88000
5	5	UAE	7	Python	Rust	Grok	PUT	90000

Operation 11

DELETE FROM "Dev_survey"
WHERE "Experience_Years" < 5;

```
sqlite> DELETE FROM "Dev_Survey"  
...> WHERE "Experience_Years" < 5;  
sqlite>
```

Database Structure Browse Data Edit Pragmas Execute SQL								
Table: Dev_Survey								
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Paki...	10	Urdu	English	MySQL	DB	55000
2	2	NULL	15	NULL	NULL	NULL	NULL	60000
3	3	USA	12	English	Spanish	MySQL	CN	70000
4	5	UAE	7	Python	Rust	Grok	PUT	90000

Operation 12

DELETE FROM "Dev_Survey"
WHERE "Salary_USD" IS NULL;

```
sqlite> DELETE FROM "Dev_Survey"  
...> WHERE "Salary_USD" IS NULL;  
sqlite>
```

Database Structure Browse Data Edit Pragmas Execute SQL									
Table: Dev_Survey									
	Dev_ID	Country	Experience_Years	Primary_Language	Secondary_Language	Database_Used	Framework	Salary_USD	
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	
1	1	Paki...	10	Urdu	English	MySQL	DB	55000	
2	2	NULL	15	NULL	NULL	NULL	NULL	60000	
3	3	USA	12	English	Spanish	MySQL	CN	70000	
4	5	UAE	7	Python	Rust	Grok	PUT	90000	